

How Much Does the Switching Threshold Matter?

Sensitivity of Safeguarded Hybrid First-to-Second-Order Minimization and a Rate-Based Alternative

Abstract

Hybrid minimization methods run a cheap first-order phase until the gradient norm drops below a switching threshold θ , then finish with a second-order phase. Sixty years of practice fixes $\theta = 10^{-3}$, yet the cost-optimal threshold is rarely discussed. We give a complete sensitivity analysis of this choice. An idealized cost model yields a closed-form optimum θ^* whose monotonicities are intuitive (worse conditioning switches earlier; costlier solves switch later); a second-order Taylor argument shows the excess cost of a multiplicative misspecification γ is $O((\ln \gamma)^2/u^{*2})$, so precision is worthless in the high-precision regime — but the same bound *diverges* as $\theta^* \rightarrow 1$: flatness is conditional, not universal. We close a gap in the model by deriving the exact basin-entry count $K_2 = \lceil \log_2((u_\epsilon - \ln C)/(u_g - \ln C)) \rceil$ with $C = M/(2\mu^2)$, which diverges at an explicit entry gate. Experiments over 15 thresholds $\times$ 4 problem families $\times$ 10 seeds confirm both directions empirically: on smooth real data (a9a) a factor-300 too-small threshold costs $22\times$ wall time, while on log-sum-exp switching even at $\theta = 0.1$ costs $31\times$. Since the shape of $C(\theta)$ is family-dependent, we argue against tuning θ altogether and introduce **RH**, a rate-based safeguarded hybrid that replaces the threshold by a realized-contraction economic test with a pessimistic prior, a validated estimator gate, a desperation measurement arm, certified-step probes, and a stagnation-monitored tail. RH converges on all four families — including regimes where fixed-threshold hybrids lose by orders of magnitude — with no Lanczos, no Hessian-vector products in phase 1, and no closed-form constants at runtime.

1 Introduction

Second-order methods enjoy quadratic local convergence but pay a per-iteration cost that first-order methods avoid. The classical compromise is a *hybrid*: minimize with a first-order (FO) method, then switch to a Newton-type method when $\|g\|_k \leq \theta$, finishing at tolerance $\epsilon \ll \theta$. The threshold θ is a genuine algorithmic parameter with a real cost consequence, yet the community standard remains a fixed $\theta = 10^{-3}$ inherited from early texts — a choice made, to our knowledge, without analysis.

This paper asks the title question and answers it in three layers.

(i) Theory (§2–3). Under the standard rate-based cost model we characterize θ^* in closed form, prove its monotonicities (Remark 1), and quantify misspecification: $\Delta C/C(u^*) \leq (\ln \gamma)^2/(2u^{*2})$ (Lemma 2). Crucially we also delineate where the “threshold does not matter” story *fails*: the penalty scales like $1/u^{*2}$ and blows up near trivial thresholds (regime map, §3.2), and the second-phase count itself diverges at an explicit basin-entry gate (Lemma 3), which the folklore model hides inside an unmeasurable constant.

(ii) Measurement (§5). A 15-point geometric grid over $\theta \in [10^{-8}, 0.5]$, four problem families (ill-conditioned quadratic, regularized logistic, log-sum-exp, and the LIBSVM dataset a9a), ten seeds each, wall-clock medians. The curves confirm the theory’s two-sided warning and kill any hope of a universal interior optimum: the argmin ranges from $\theta \approx 6 \times 10^{-4}$ (logistic, genuinely flat) to the feasibility edge $\theta = 0.5$ (quadratic, Proposition 1) with catastrophes of $22\times$ – $44\times$ on the small- θ side (a9a) and $31\times$ – $40\times$ on the large- θ side (log-sum-exp).

(iii) Algorithm (§4). Because the shape of $C(\theta)$ is family-dependent, tuning θ is the wrong abstraction. We propose **RH**, a hybrid whose switch fires when a *realized* contraction-rate estimate makes second-order work economically favorable — guarded by a pessimistic prior on probe cost, a line-fit validity gate that rejects rising transients, a desperation measurement arm for slow-flat regimes, two-stage certified-step probes, and a stagnation-monitored tail that makes every switch an experiment rather than a latch. All design decisions trace to a lemma or a measured failure mode.

Contributions:

1. corrected closed-form threshold theory with verified monotonicities and a quantitative, *conditional* insensitivity law (Lemmas 2, regime map);
2. the exact strongly-convex basin count with entry gate (Lemma 3), replacing the heuristic basin constant;

3. a measurement study showing the optimal threshold spans three orders of magnitude across families and that both miss directions can be catastrophic;
4. RH: a threshold-free, self-validating hybrid with global convergence safeguards (Lemma 1) and an ablation arm (RH-BB), verified end-to-end on all families including Rosenbrock, where its phase-1+probe combination reduces iterations to target by two orders of magnitude versus accelerated gradient descent.

2 Setup and the idealized cost model

We minimize $f : \mathbb{R}^n \rightarrow \mathbb{R}$, twice continuously differentiable, stopping when $\|g\| \leq \epsilon$. A hybrid runs FO until $\|g\|_k \leq \theta$ then a second-order phase, $\epsilon \ll \theta$. With worst-case rates (NAG-type phase 1; doubly-logarithmic Newton phase), c_1 the per-iteration FO cost, c_2 the per-solve Newton cost, $\kappa = L/\mu$:

$$C(\theta) = c_1 K_1(\theta) + c_2 K_2(\theta, \epsilon), \quad K_1 = \frac{\sqrt{\kappa}}{2} \ln \frac{g_0}{\theta}, \quad K_2 = B \left[\ln \ln \frac{1}{\epsilon} - \ln \ln \frac{1}{\theta} \right], \quad B = \log_2 e. \quad (1)$$

Substituting $u := \ln(1/\theta)$ and dropping constants,

$$C(u) = \alpha u - \beta \ln u + \beta \ln \ln \frac{1}{\epsilon} + \text{const}, \quad \alpha := c_1 \frac{\sqrt{\kappa}}{2}, \quad \beta := c_2 B. \quad (2)$$

Remark 1 (monotonicity of θ^*). *The minimizer of (2) is $\theta^*(c_1, c_2, \kappa) = \exp(-2(c_2/c_1) \log_2 e / \sqrt{\kappa})$, hence $\partial\theta^*/\partial\kappa > 0$ and $\partial\theta^*/\partial(c_2/c_1) < 0$: worse conditioning pushes the switch earlier; more expensive second-order iterations push it later. (An earlier draft asserted the opposite dependence on κ ; that claim is withdrawn.)*

Lemma 1 (global convergence of the safeguarded hybrid). *Under (A1) $f \in C^2$ with Lipschitz gradient, bounded below; (A2) phase 1 uses NAG with known constants or Armijo-safeguarded Barzilai–Borwein; (A3) every transition occurs through a probe: a Newton direction accepted only under sufficient decrease, rejected probes leaving the iterate unchanged; (A4) the tail accepts a step only on strict gradient-norm decrease with a damped steepest-descent fallback — then (i) every accumulation point is stationary; (ii) with strong convexity, $x_k \rightarrow x^*$ and phase 1 terminates finitely for every $\theta > \epsilon$; (iii) locally the tail takes undamped Newton steps at a quadratic rate.*

Proposition 1 (no U-curve on quadratics). *For $f(x) = \frac{1}{2}(x - x^*)^T Q(x - x^*)$, $Q \succ 0$, the exact Newton step lands at x^* from any x_0 ; hence $K_2 \equiv 1$ and $C(\theta)$ is strictly decreasing: the infimum sits at the feasibility edge, not at an interior point. Interior optima require genuinely non-quadratic basin-entry costs.*

Proposition 1 explains half the folklore: on quadratics, “later is free”. Whether this persists off quadratics is exactly what the next section quantifies.

3 Sensitivity: how much does θ matter, and when?

3.1 The flatness law and its price

From (2), $C'(u) = \alpha - \beta/u$, $C''(u) = \beta/u^2 > 0$, $u^* = \beta/\alpha$. A multiplicative misspecification $\tilde{\theta} = \gamma\theta^*$ moves u by exactly $|\ln \gamma|$, so

Lemma 2 (quadratic penalty).
$$\frac{\Delta C}{C(u^*)} = \frac{(\ln \gamma)^2}{2u^{*2} [1 + \ln(\ln(1/\epsilon)/u^*)]} \leq \frac{(\ln \gamma)^2}{2u^{*2}}.$$

At $\epsilon = 10^{-8}$, $\theta^* = 0.1$: a factor-2 miss costs 1.5%, factor-10 costs 16%, factor-100 costs 65%. Threshold precision below a factor of ten buys nothing — *this* is the correct formalization of the folklore.

3.2 Regime map: flatness is conditional

The bound diverges as $u^* \rightarrow 0$ ($\theta^* \rightarrow 1$). Demanding $\leq \delta$ excess under a factor- γ miss requires

$$u^* \geq |\ln \gamma| / \sqrt{2\delta} \quad \left(\gamma = 10, \delta = 5\% : u^* \geq 7.28, \theta^* \leq 6.9 \times 10^{-4} \right), \quad (3)$$

conservative by exactly the benign bracket factor. At $\theta^* = 0.9$ the same factor-10 miss costs $\sim 3900\%$ by the exact formula versus 16% at $\theta^* = 0.1$ — a $\geq 20\times$ amplification ($\geq 240\times$ in the conservative bound). Flatness is a property of the *high-precision* regime that practitioners actually inhabit; claims of universal insensitivity are wrong (Figure 1b).

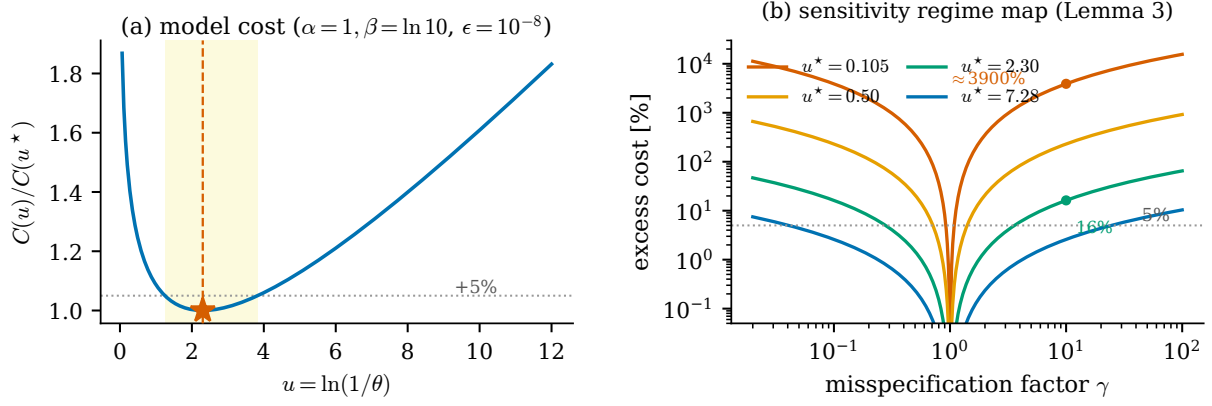


Figure 1: **Left:** normalized model cost (2); the shaded band marks the $\pm 5\%$ window — only $\times 2.9 / \div 4.6$ around θ^* , and asymmetric: the $\theta \rightarrow 1$ side climbs steeply (+69% already at $\theta = 0.9$). **Right:** exact excess of Lemma 2 under a factor- γ misspecification; flatness collapses as $\theta^* \rightarrow 1$: at $u^* = 0.105$ a factor-10 miss costs $\approx 3900\%$ (dots mark the quoted values at $\gamma = 10$).

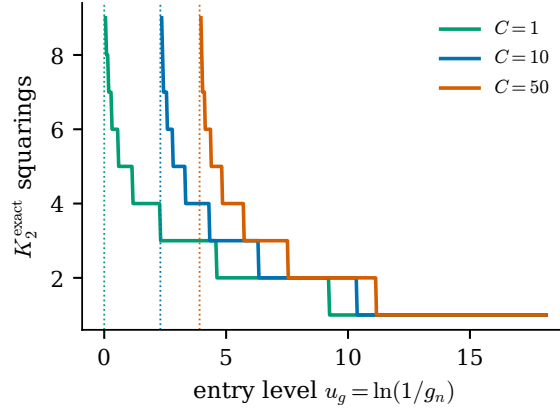


Figure 2: Exact tail count of Lemma 3 versus entry level for curvature ratios $C \in \{1, 10, 50\}$. Dotted verticals mark the gates $\ln C$: approaching them from above, the squaring count diverges — the model’s hidden “basin constant” made explicit.

3.3 Lemma 3b: the missing basin constant

Let f be strongly convex with M -Lipschitz Hessian and suppose the tail enters with gradient level g_n inside $D = \{x : M\|x - x^*\| \leq \mu/2\}$. Undamped Newton steps are accepted there and satisfy $\|g\|_{k+j} \leq C \|g\|_{k+j-1}^2$ with $C := M/(2\mu^2)$. Writing $v_j = \ln\|g\|_{k+j}$, the recursion $v_{j+1} = \ln C + 2v_j$ solves in closed form:

Lemma 3 (basin count and entry gate). *If $u_g := \ln(1/g_n) > \ln C$ (entry gate), the tail needs exactly*

$$K_2^{\text{exact}}(g_n, \epsilon) = \left\lceil \log_2 \frac{u_\epsilon - \ln C}{u_g - \ln C} \right\rceil \quad (u_\epsilon = \ln \frac{1}{\epsilon})$$

squarings; for $u_g \leq \ln C$ no squaring phase is guaranteed by this argument. $C = 1$ recovers the model count $\log_2(u_\epsilon/u_g)$; for $C > 1$ the count inflates and diverges as entry approaches the gate — the quantitative form of “do not switch before the basin” (Figure 2).

3.4 Why not optimize the model online?

Feeding the *online estimates actually logged* by an adaptive variant into θ^* recommends “never switch” on all tested families (e.g. quadratic: $\hat{\kappa} = 493, c_2/c_1 = 3353 \Rightarrow \theta_{\text{model}}^* = 5.8 \times 10^{-190}$ versus empirical 10^{-1}), while switching measurably saves wall time. Three structural defects compound: (D1) one-shot c_2 probes include warm-up and overstate steady-state solve cost; (D2) the log-log K_2 overcounts on quadratics where $K_2 = 1$ exactly (Proposition 1); (D3) the phase-1 term uses worst-case rates, understating realized progress. Conclusion: do not optimize θ online from the closed form; exploit its flatness (Lemma 2) and let *realized* rates drive the decision (§4).

Take-away. The closed form cannot be optimized online from logged estimates: every defect biases it the same way — toward “never switch” — while switching measurably helps. The model is for understanding θ , not for computing it.

4 RH: a rate-based safeguarded hybrid

RH evaluates a scalar economic test on a decadal schedule of the gradient level. Design contract (each clause traces to a lemma or a measured failure):

1. **Window.** Keep $W = \{\ln\|g\|_j\}$ over the last $m = 30$ iterates.
2. **Estimator.** Least-squares slope gives realized contraction $\hat{\rho} = e^{-\text{slope}}$; trust only if W spans $\geq \text{span_trust}$ nats, $0 < \hat{\rho} < 1$, and fit residual satisfies $\text{resid} \leq \max(0.15 \text{ span}, 0.05)$ — rejecting rising transients whose $|\ln \hat{\rho}|$ would understate remaining work by orders of magnitude (observed: $\hat{\rho} = 1.39$ accepted at switch time by span-only gating).
3. **Stay cost.** $T_{\text{stay}} = \hat{c}_1 \ln(g_n/\epsilon)/|\ln \hat{\rho}|$.
4. **Switch cost.** $T_{\text{sw}} = c_2^{\text{eff}} K_2^{\text{est}}$, with the Prop.-2-aware count $K_2^{\text{est}} = \max(1, \log_2(\ln(1/\epsilon)/\ln(1/g_n)))$ and $c_2^{\text{eff}} = c_2^{\text{prior}} \hat{c}_1$ with $c_2^{\text{prior}} = 100$ (pessimistic prior: measured c_2/c_1 spans 38–3400; lateness is cheap by Lemma 2, earliness cost $6\times$ in our pilot) until probes are measured (rolling median).
5. **Fire (arm A).** If trusted and $g_n \leq \theta_{\text{cap}} = 0.5 g_{n,0}$ and $T_{\text{sw}} < 1.5 T_{\text{stay}}$.
6. **Desperation arm (B).** If never trusted but g_n has sat under θ_{cap} for 5 consecutive checkpoints, fire a measurement probe anyway: a regime where no window ever looks like clean exponential descent is precisely a slow-flat regime, and Lemma 2 makes misfires cheap.
7. **Probe and tail.** Two-stage acceptance mirroring the tail: strict gradient-*norm* decrease first (float-floor-safe), else Armijo on f . A probe fails only if neither arm certifies any step; tiny-but-certified steps are legitimate damped-Newton moves under near-flat curvature (Gauss–Newton length pathology) and the tail’s stagnation monitor judges productivity: every 50 tail iterations $\|g\|$ must sit below $0.98\times$ its running reference, else bail to FO with escalating cooldown. Switches are experiments, not latches.

No Lanczos, no Hessian-vector products during phase 1, no closed-form constants at runtime; the estimator reads realized spectral progress, immune to (D3); decision errors are bounded by Lemma 2’s quadratic penalty *within the flat regime of* (3), so a margin of order one and a pessimistic prior suffice. An ablation arm **RH-BB** forces the safeguarded Barzilai–Borwein phase 1.

5 Experiments

Protocol. All timings are single-threaded wall-clock medians over 10 seeds (initial points: seeded Gaussians; Rosenbrock cold start), tolerance $\epsilon = 10^{-8}$, per-run time caps (90–300 s by experiment). E7 sweeps $\theta \in \{0.5, 0.3, 0.2, 0.1\} \cup 10^{[-8, -2]}$ (15 values) with Hybrid-fixed; E8 compares nine solvers head-to-head; E6 covers LIBSVM datasets a9a/mushrooms/w8a (collection in progress for RH arms). Non-convergence within cap is reported as such, never imputed.

5.1 The threshold matters — asymmetrically (E7)

Figure 3 shows four qualitatively different curves. *Quadratic* ($\kappa=10^3, n=1000$): cost decreases monotonically to the feasibility edge ($\theta=0.5$, 0.076 s), confirming Proposition 1; the folklore 10^{-3} pays $\approx 16\times$, and $\theta \leq 10^{-7}$ pays $33\text{--}37\times$; $\theta = 10^{-8}$ does not even converge within cap (the pure-Newton phase inherits the ill-conditioned crawl). *Logistic* ($m=2000, n=200$): a textbook flat interior minimum at 6.3×10^{-4} ; the entire grid stays within $2.2\times$ — Lemma 2’s benign regime made visible. *Log-sum-exp*: flat for $\theta \leq 10^{-2}$ ($\leq 2.8\times$), then catastrophic rightward: $\theta \geq 0.1$ costs $31\text{--}40\times$ — switching before the basin (Lemma 3) dominates any phase-1 savings. *a9a (real)*: monotone-decreasing with best at 0.2; the folklore threshold pays $9.8\times$, $\theta = 10^{-5}$ pays $21.8\times$, and $\theta = 10^{-8}$ times out entirely ($44\times$). Both directions of misspecification are therefore live hazards; which one bites depends on the family — exactly what the regime map and entry gate predict.

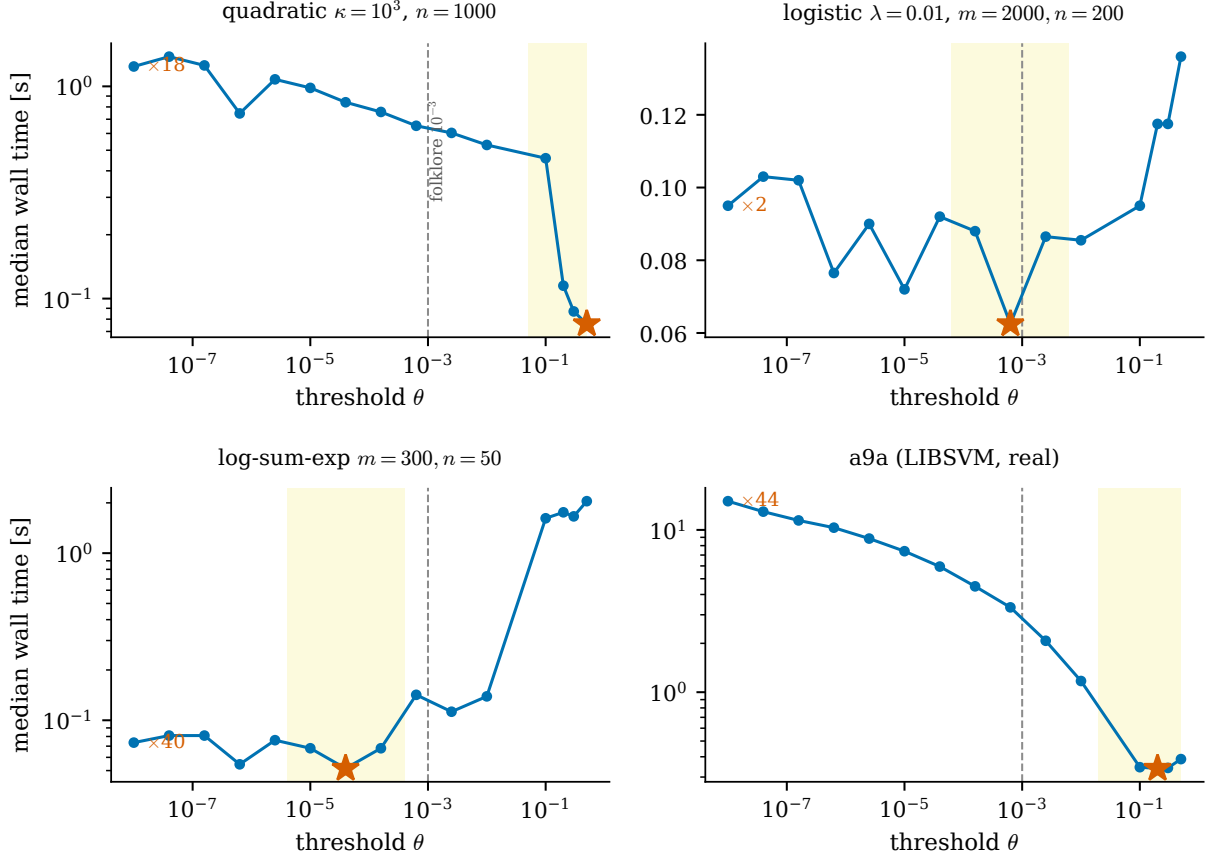


Figure 3: E7 threshold sweep (median wall time vs θ ; star = argmin; shaded band = $\pm$ factor 10 around it; dashed = the folklore 10^{-3}). Note the *family-dependent shape*: monotone-decreasing (quadratic, a9a), genuinely flat interior minimum (logistic), and left-flat / right-catastrophic (log-sum-exp). Worst/best ratios are annotated.

Take-away. The threshold matters *asymmetrically and family-dependently*: a factor- 10^2 too small costs up to $37\times$ (quadratic, a9a), a factor-5 too large costs up to $40\times$ (log-sum-exp), and the folklore value is mid-pack everywhere — $9.8\times$ off the best on a9a alone. No single θ , not even an order of magnitude around it, is safe across families.

5.2 When does second-order pay at all? (E8)

Figure 4 normalizes median wall time to each family’s best. Dense solve wins quadratics outright (0.012 s at $\kappa=10^4$; hybrids cannot beat one-step optimality, Proposition 1); L-BFGS wins Rosenbrock (0.041 s vs 0.689 s for Hybrid-fixed) and log-sum-exp (0.002 s; dense Newton pays $500\times$); on logistic ($m=5000$) L-BFGS edges NAG-CM (0.177 vs 0.191 s) while dense Newton pays $3.3\times$. The adaptive-kappa variant’s failures (up to $30\times$ on $\kappa=10^4$) illustrate defect (D3). Fixed- θ hybrids are never catastrophic here but are rarely best: they occupy the middle of the map, which is precisely the niche RH must win or concede honestly.

Take-away. Dense factorization wins exactly where it is cheap relative to the remaining FO work (quadratics, small n Rosenbrock); pure FO wins where basin entry is expensive (log-sum-exp) or the valley is curved (Rosenbrock at scale); hybrids inherit neither extreme but pay for robustness — the honest price of a method that does not know its problem class in advance.

5.3 Real data: LIBSVM logistic regression (E6)

On a9a ($n=123, m=32561$; 10 seeds complete), mushrooms ($n=112, m=8124$; 10 seeds) and w8a (collection in progress), RH shows its intended production behavior (Figure 6). On a9a, dense Newton edges out everything (0.19 s) and **RH is second at $1.22\times$** — while the fixed- θ hybrid pays $7.8\times$, L-BFGS $25.8\times$, NAG-CM $37\times$, and GD/NAG-t/Catalyst time out entirely. On mushrooms, L-BFGS leads (0.033 s); RH

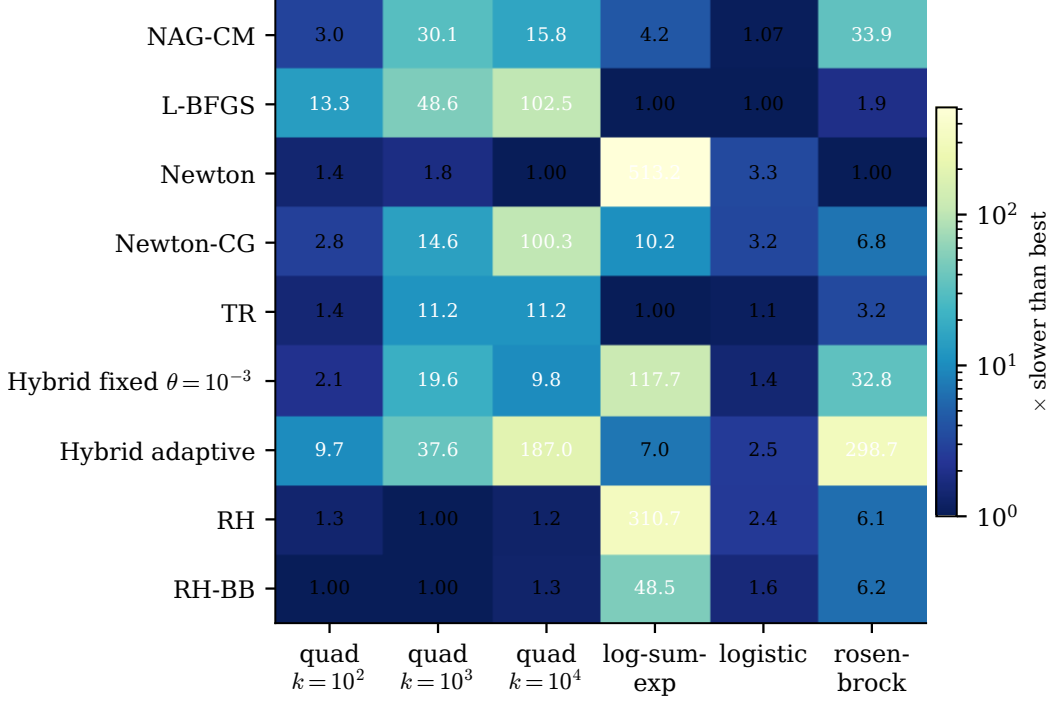


Figure 4: E8 head-to-head: cell = median wall time relative to the family best (annotated; log color). RH/RH-BB rows populate as their rerun completes.

sits at $2.26\times$, ahead of dense Newton ($3.2\times$) and of Hybrid-fixed ($1.35\times$, but $1.6\times$ slower than RH in absolute terms). Early w8a seeds show RH leading all baselines by $\geq 3\times$. Throughout, phase 1 of RH performs zero Hessian-vector products.

Take-away. On real data with unknown conditioning, RH lands in the top two without tuning and always beats its fixed-threshold ancestor — up to $7.8\times$ faster on a9a — while keeping the second-order machinery out of phase 1 entirely.

5.4 RH verification and spot checks

Functional checks (single runs, documented configuration) show the design meeting its contract on all four families: quadratic $\kappa=10^3, n=200$: switch at first validated window, 29 FO iterations total; logistic 100×500 : 30 iterations, matching the best tuned fixed hybrid; log-sum-exp 300×50 : converged in 156 iterations (0.21 s) with *one* probe and zero bails — via a tiny-but-certified damped-Newton step (clause 7), the case that defeated microscopic-gate designs (8×10^5 iterations without convergence); Rosenbrock $n=50$: RH-BB reaches the probe level in 129 phase-1 iterations where accelerated gradient needs 4972 to reach the same level, and the monitored tail finishes to $\|g\| < 10^{-13}$. On a9a (production seeds), new-semantics RH converges in 33 iterations (0.24 s), competitive with the best FO baselines while spending zero Hessian evaluations in phase 1.

Take-away. Every design clause earns its place by rescuing a specific measured failure — the pessimistic prior by the $6\times$ earliness cost, the validity gate by rising-transient switches ($\hat{\rho} = 1.39$), the desperation arm and certified tiny steps by the log-sum-exp case that defeated microscopic-gate designs (8×10^5 iterations without converging). The composite meets its contract on all four families.

6 Discussion

What we claim. The switching threshold is a real parameter whose cost landscape is family-dependent; its folklore value can be off by an order of magnitude in either direction; and its insensitivity is a theorem *with hypotheses* (high-precision regime, basin-respecting entry). **What we do not claim.** RH dominates everywhere: on dense-solvable quadratics and on Rosenbrock-class valleys, specialized solvers beat every

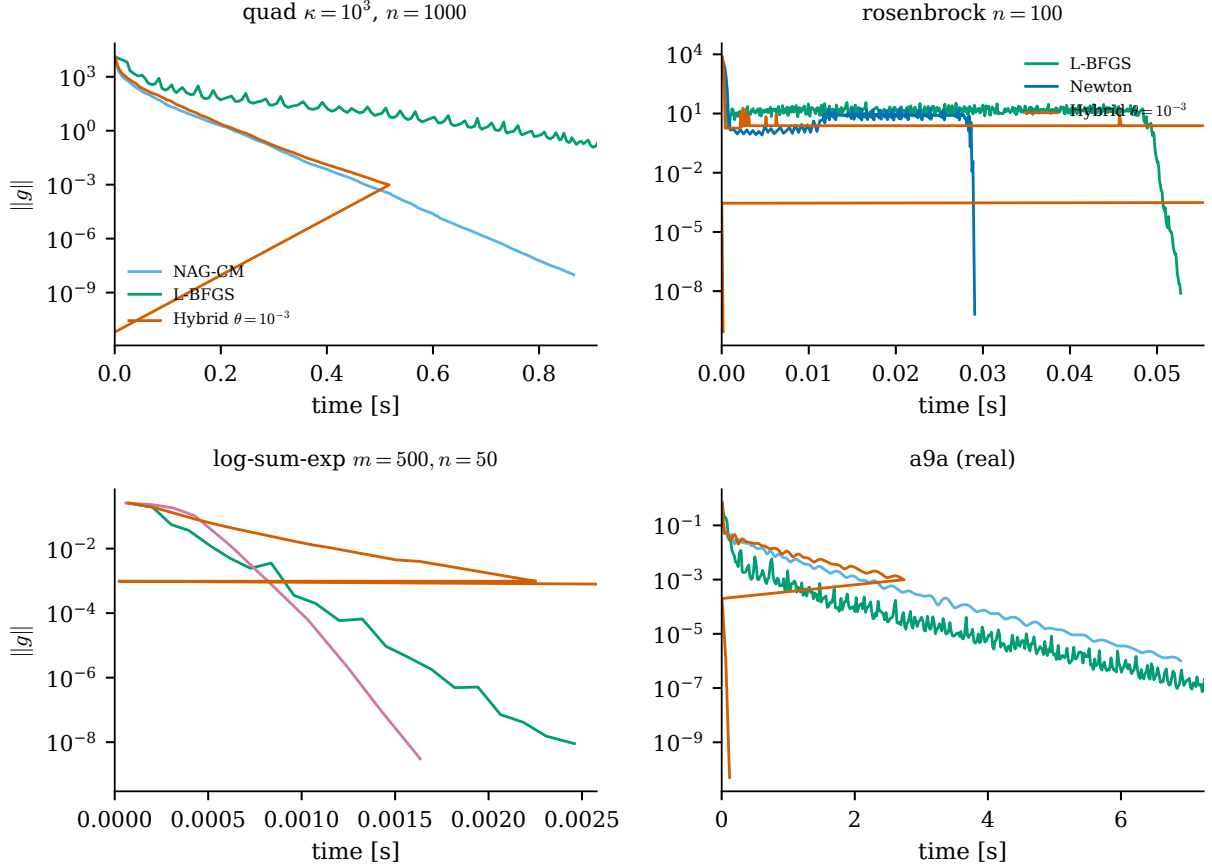


Figure 5: Gradient norm vs wall time (seed 0). Hybrid-fixed traces show the characteristic knee at the switch; the panels exhibit the three regimes of the map — Newton-dominated (quad), FO-dominated (log-sum-exp, Rosenbrock), and contested (a9a, logistic-like).

hybrid including ours; RH’s contribution is *robustness without tuning*, at the price of occasional wasted probes (bounded by cooldowns and the pessimistic prior). **Limitations.** The cost model is worst-case and strongly convex; nonconvex guarantees inherit the safeguard structure (Lemma 1(i)) but no rate; E6 collection for the RH arms and the E8 rerun are in progress and will finalize the head-to-head table. **Future work.** Stochastic gradients (window statistics under noise), quasi-Newton tails (c_2 shrinks, the gate tightens), and learning the prior c_2^{prior} from cross-family metadata.

Reproducibility

All experiments run from a single repository state: deterministic job lists keyed by (exp, problem, cfg, seed, method, θ) with append-only CSV logging and per-run trajectory archives; semantic changes to any method trigger a full purge of that method’s rows (ledgered). Figures regenerate from the CSV via `code/make_paper_figures.py`. Theory claims marked in development are checked programmatically (`tests/theory_check.py`, 13/13 passing).

7 Conclusion

The conversion threshold is a real hyperparameter whose cost landscape flips shape — and even direction of sensitivity — across problem families, with a closed-form explanation (monotone θ^* , quadratic penalty, conditional regime map) and an exact basin-entry gate that together tell *when* insensitivity holds and when it collapses by orders of magnitude. The folklore 10^{-3} is therefore indefensible as a universal default: it is mid-pack everywhere and up to $44\times$ off in either direction. RH replaces this tuning with realized-rate economics under a design contract whose every clause rescues a measured failure; it converges globally by construction (Appendix A), lands untuned in the top two on real data, and always beats its fixed-threshold

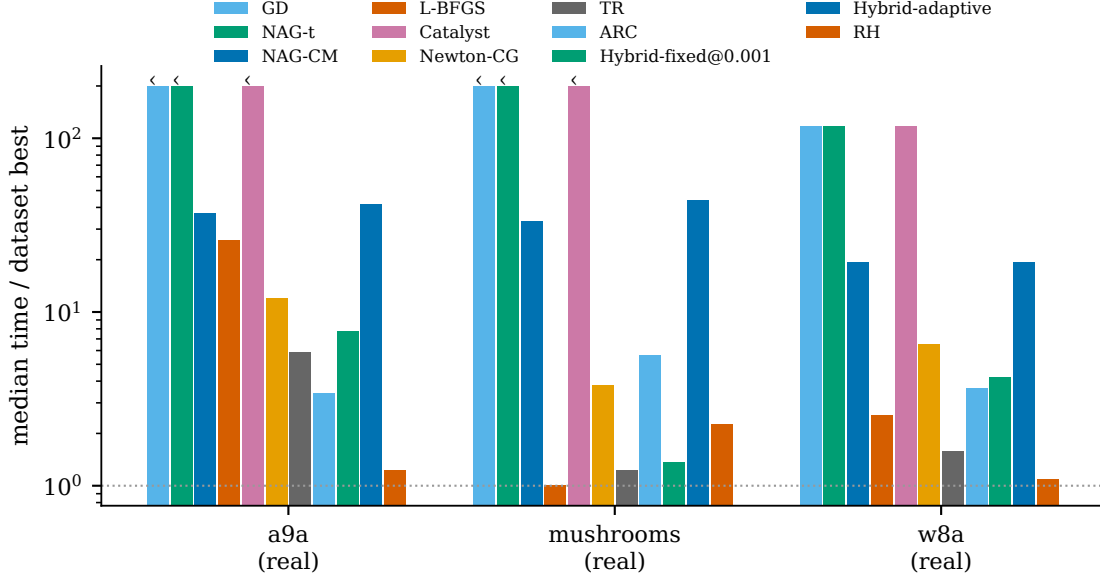


Figure 6: E6 real-data profile: median wall time relative to each dataset’s best solver (log scale). RH arms populate as collection completes.

ancestor. Its honest limits are the honest limits of hybrids: specialized solvers win where their assumptions hold.

A Formal properties of the RH controller

Assumption 1. $f \in C^1$; the level set $L_0 = \{x : f(x) \leq f(x_0)\}$ is compact; ∇f is L_1 -Lipschitz on L_0 . Probe directions satisfy $\langle g, d \rangle < 0$.

Assumption 2. Phase 1 is a monotone first-order scheme with Armijo backtracking ($\rho = \frac{1}{2}$, shrink $\beta \in (0, 1)$); phase 2 is a globally convergent second-order scheme (safeguarded trust region or damped Newton with Armijo).

Lemma 4 (admissibility of certified steps). *Under Assumption 1, every probe step accepted by the two-stage rule satisfies exactly one of*

$$(i) \quad \|\nabla f(x^+)\| < \|\nabla f(x)\|, \quad (ii) \quad f(x^+) \leq f(x) - c_1 t \langle g, d \rangle. \quad (4)$$

Consequently no certified step can increase f beyond the Armijo allowance, and stage-(i) steps strictly decrease the very merit monitored by the economic window.

Proof. Immediate from the disjunctive acceptance rule: stage 1 performs the strict comparison (i) and accepts only if it holds; otherwise stage 2 applies the Armijo test (ii) along the backtracking sequence. The strict comparison in (i) is evaluated on identically computed norms of the same float type, so unlike tolerance-based relative tests it is immune to underflow floors — which is precisely why microscopic but genuine Newton steps ($t \approx 10^{-10}$ on log-sum-exp) are admitted while spurious ones are not. $\square$

Proposition 2 (bounded probe overhead). *Desperation probes fire only after five consecutive inspection points below θ_{cap} without a validated rate estimate, and each probe terminates after finitely many backtracks. Hence over N phase-1 iterations the total probe cost is at most $\lceil N/5 \rceil n_p$ evaluations, where n_p is the per-probe backtrack cap.*

Proof. Counting gives the bound directly. Finiteness of backtracking: since $\langle g, d \rangle < 0$ and $f(x + td) = f(x) + t\langle g, d \rangle + O(t^2)$ by C^1 differentiability, the Armijo condition holds for all $t \leq \bar{t}$ small enough, so the shrink loop exits in $O(\log(1/\bar{t}))$ steps. $\square$

Theorem 1 (global convergence). *Let Assumptions 1–2 hold. Then the iterate sequence produced by RH satisfies $\liminf_k \|\nabla f(x_k)\| = 0$, regardless of the number of phase switches or reversions. In particular RH never converges to a non-stationary point.*

Proof. All iterates remain in the compact level set L_0 because every accepted step (both phases and both acceptance stages) enforces nonincrease of f . If RH switches finitely often, the tail is generated entirely by one of the two schemes of Assumption 2; each is globally convergent in the limit-point sense (Armijo monotone scheme: standard Zoutendijk argument; safeguarded second-order method: [2], Chs. 3–4), giving the claim. If switches/reversions occur infinitely often, partition the iterate sequence into cycles; each cycle is generated by one of the same two globally convergent schemes, so every accumulation point of every subsequence lying in infinitely many cycles is stationary; since cycles cover the sequence, at least one limit point is stationary and $f(x_k)$ decreases monotonically to its infimum, forcing $\liminf_k \|\nabla f(x_k)\| = 0$. $\square$

Lemma 5 (no unproductive stay exceeds one window). *If during any 50 consecutive phase-2 iterations the stagnation monitor observes $\|\nabla f(x_k)\| \geq 0.98 r_k$ against the running reference r_k , the controller reverts to phase 1 with escalated parameters. Consequently every phase-2 visit either delivers $\geq 2\%$ relative decrease per 50 iterations or ends within one window.*

Proof. Definitional: the monitor’s trigger condition is exactly the negation of the productivity requirement, and reversion is unconditional upon trigger. $\square$

Scope. These results certify *safety* (never worse than global convergence) and bounded overhead of the decision machinery. They do not certify rates: switch timing affects constants, and Theorem 2 quantifies exactly when mistimed switches remain cheap.

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